MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 08** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. A circular disc of radius 10 cm is divided into 8 equal sectors. The central angle of each sector is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $75°$

Q2. A square of side 10 cm has an incircle of radius r and a circumcircle of radius R. The ratio of their areas is: [1]

(A) $1 : 2$ (B) $1 : \sqrt{2}$ (C) $1 : 4$ (D) $2 : 3$

Q3. In a square $ABCD$ of side 7 cm, two quadrants are drawn with centres A and C and radius 7 cm. The area of the shaded leaf region between them is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 28 cm^2 (B) 14 cm^2 (C) 35 cm^2 (D) 21 cm^2

Q4. An automobile has two wipers of length 21 cm each, sweeping through an angle of $120°$. If the blades do not overlap, the total area swept at each sweep is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 462 cm^2 (B) 924 cm^2 (C) 308 cm^2 (D) 616 cm^2

Q5. In a circle of radius 6 cm, a chord subtends an angle of $60°$ at the centre. The area of the corresponding equilateral triangle formed by the radii and chord is: [1]

(A) $9\sqrt{3}$ cm^2 (B) $18\sqrt{3}$ cm^2 (C) $36\sqrt{3}$ cm^2 (D) $6\sqrt{3}$ cm^2

Q6. **Assertion (A):** If the radius of a circle is doubled, its area becomes four times its original area.
Reason (R): The area of a circle varies directly as the square of its radius ($A = \pi r^2$). [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. A bicycle wheel of diameter 70 cm rotates at 250 revolutions per minute. Find the speed of the cyclist in km/h. $\left(\text{Use } \pi = \dfrac{22}{7}\right)$ [2]

Q8. In a circle of radius 21 cm, an arc subtends an angle of $60°$ at the centre. Find the length of the arc. $\left(\text{Use } \pi = \dfrac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. The length of the minute hand of a wall clock is 14 cm. Find the area swept by the minute hand in 20 minutes. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

Q10. A circular field has a circumference of 660 m. A square sports plot of the largest possible area is inscribed in this field. Find the area of the square sports plot. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

Q11. A chord of a circle of radius 15 cm subtends an angle of 60° at the centre.

(a) Find the area of the corresponding minor segment of the circle.

(b) Find the area of the corresponding major segment of the circle.

(c) Determine the ratio of the area of the minor segment to that of the major segment. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$)

[5]

Q12. Two circular flower beds are established on two opposite sides of a square lawn $ABCD$ of side 56 m. The centre of each circular flower bed is the intersection point O of the diagonals of the lawn.

(a) Find the radius of each circular sector.

(b) Calculate the total area of the entire garden (lawn + flower beds) using the sector method.

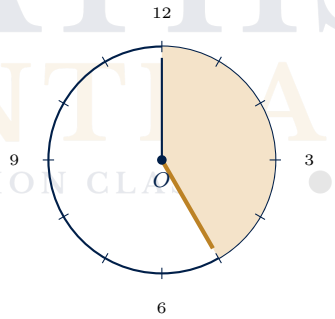
(c) Verify the result by summing the area of the square and the two circular segment beds. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Antique Grandfather Clock Rotational Geometry

A horologist studies an antique grandfather clock. The minute hand has length 14 cm and the hour hand has length 7 cm. Both hands rotate smoothly about the common dial centre O.



(i) Calculate the angle through which the minute hand rotates in 25 minutes. [1]

(ii) Calculate the area swept by the minute hand between 8:00 am and 8:25 am. [1]

(iii) Calculate the area swept by the hour hand in 1 hour, OR find the distance traversed by the tip of the minute hand in 45 minutes. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★